LESSON PLAN

AIRCRAFT HUSBANDRY, CORROSION PREVENTION AND SURFACE PROTECTION (27)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to mixed explosives and their advantages, Application
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to mixed explosives and their advantages, Application
5. Specific Objectives	27.1.1.1 Trainee will be able to explain about various explosives

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of explosive safety cards and identification of mixed explosive labels on storage containers
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary difference between single-base and mixed explosives?
(b) Why are mixed explosives preferred over single-base explosives in modern aviation applications?
(c) Can you list three common applications of mixed explosives in aircraft maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of mixed explosives • Composition of oxidizer and fuel components • Advantages of mixed explosives • Stability and safety characteristics • Common types of mixed explosives	L & Ds	PPT, IB	15 min
(b)	• Application in aircraft ordnance • Usage in pyrotechnic devices • Role in emergency systems • Handling and storage requirements • Safety precautions during application	L & Ds	PPT, PDF	20 min
(c)	• Comparison with single-base explosives • Environmental impact considerations • Future trends in explosive technology • Regulatory compliance for mixed explosives • Summary of key advantages	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the primary advantage of using mixed explosives over single-base explosives in aircraft systems.
(b) Describe the specific safety precautions required when handling mixed explosives in an aircraft hangar.

12. Home Assignment and Follow Up	Prepare a brief report on the specific application of mixed explosives in emergency ejection systems, referencing the next lesson plan (LP 02) on advanced pyrotechnic devices.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Senior Instructor, P No. 12345	Senior Officer, Chief Instructor, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min